Models:

- 3 models with merged data (10k, 60k, full)
- 2 models with full from each (companies, movies)

Entity set:

- Wikidata

Batch size:

- 256

1. merged_10k

- Consists of merged data from both ‘movies_test.csv’ og ‘companies_test.csv’
- Uses 5000 lines from both files (total 10.000)
- roughly ~15 min training time (Batch 128, 4070ti)
- Results:

Companies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.7750   77.50   (775/1000)
Precision: 0.9639
Recall:    0.7750
F1 Score:  0.8592
================================
```

```
f1: 0.4015
accuracy: 0.7730
gold_recall: 0.9220
p: 0.2711
r: 0.7730
num_gold_spans: 1000
```

- Increase from 72% to 77.5% **(+5.4%)**
- Refined accuracy: 77.3% **(+5.3%)**

Movies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.8860   88.60   (886/1000)
Precision: 0.9683
Recall:    0.8860
F1 Score:  0.9253
================================
```

```
f1: 0.3848
accuracy: 0.8860
gold_recall: 0.9090
p: 0.2458
r: 0.8860
num_gold_spans: 1000
```

- Increase from 70.2% to 88.6% **(+18.4%)**
- Refined accuracy: 88.6% **(+18.4%)**

2. merged_60k

- Consists of merged data from both ‘movies_test.csv’ og ‘companies_test.csv’
- Uses 30.000 lines from both files (total 60.000)
- roughly 1 hour 37 minutes (Batch 128, 4070ti)
- Results:

Companies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.7780   77.80   (778/1000)
Precision: 0.9798
Recall:    0.7780
F1 Score:  0.8673
==============================
```

```
f1: 0.4486
accuracy: 0.7810
gold_recall: 0.9220
p: 0.3147
r: 0.7810
num_gold_spans: 1000
```

- Increase from 72% to 77.8% **(+5.6%)**
- Refined accuracy: 78.10% **(+6.1%)**

Movies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.8920   89.20   (892/1000)
Precision: 0.9791
Recall:    0.8920
F1 Score:  0.9335
==============================
```

```
f1: 0.4592
accuracy: 0.8920
gold_recall: 0.9090
p: 0.3092
r: 0.8920
num_gold_spans: 1000
```

- Increase from 70.2% to 89.2% **(+19%)**
- Refined accuracy: 89.2% **(+19%)**

3. companies_full

- Consists of full data from only ‘companies_test.csv’
- Uses 67.500 lines
- roughly 1 hour 34 minutes (Batch 128, 4070ti)
- Results:
-

Companies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.7900   79.00   (790/1000)
Precision: 0.9887
Recall:    0.7900
F1 Score:  0.8783
==============================
```

```
f1: 0.4510
accuracy: 0.7890
gold_recall: 0.9220
p: 0.3157
r: 0.7890
num_gold_spans: 1000
```

- Increase from 72% to 79% **(+7.0%)**
- Refined accuracy: 78.9% **(+6.9%)**

Movies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.3830   38.30   (383/1000)
Precision: 0.8201
Recall:    0.3830
F1 Score:  0.5222
==============================
```

```
f1: 0.1624
accuracy: 0.3940
gold_recall: 0.9090
p: 0.1023
r: 0.3940
num_gold_spans: 1000
```

- Decrease from 70.2% to 38.3% **(-31.9%)**
- Refined accuracy: 39.4% **(-30.8%)**

4. movies_full

- Consists of full data from only ‘movies_test.csv’
- Uses 67.500 lines
- roughly 1 hour 21 minutes (Batch 128, 4070ti)
- Results:

Companies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.0790   7.90   (79/1000)
Precision: 0.8876
Recall:    0.0790
F1 Score:  0.1451
================================
```

```
f1: 0.0866
accuracy: 0.0970
gold_recall: 0.9220
p: 0.0783
r: 0.0970
num_gold_spans: 1000
```

- decrease from 72% to 7.9% **(-64.1%)**
- Refined accuracy: 9.7% **(-62,3%)**

Movies:

Custom accuracy results:

```
========-EVAL METRICS-==========
Accuracy:  0.8930   89.30   (893/1000)
Precision: 0.9900
Recall:    0.8930
F1 Score:  0.9390
================================
```

```
f1: 0.4815
accuracy: 0.8920
gold_recall: 0.9090
p: 0.3298
r: 0.8920
num_gold_spans: 1000
```

- Increase from 70.2% to 89.3% **(+19,1%)**
- Refined accuracy: 89.2% **(+19.0%)**

5. merged_full

- Consists of full data from both ‘movies_test.csv’ and ‘companies_test.csv’
- Uses (67.500 x 2) = 135.000 lines
- roughly 3 hour 13 min (Batch 128, 4070ti)
- Results:

Companies:

Custom accuracy results:

```
========-EVAL METRICS-=========
Accuracy:  0.7860   78.60   (786/1000)
Precision: 0.9924
Recall:    0.7860
F1 Score:  0.8772
==============================
```

```
f1: 0.4488
accuracy: 0.7840
gold_recall: 0.9220
p: 0.3144
r: 0.7840
num_gold_spans: 1000
```

- Increase from 72% to 78.6% **(+6.6%)**
- Refined accuracy: 78.4% **(+6.4%)**

Movies:

Custom accuracy results:

```
========-EVAL METRICS-=========
Accuracy:  0.8920   89.20   (892/1000)
Precision: 0.9966
Recall:    0.8920
F1 Score:  0.9414
==============================
```

```
f1: 0.4195
accuracy: 0.8920
gold_recall: 0.9090
p: 0.2742
r: 0.8920
num_gold_spans: 1000
```

- Increase from 70.2% to 89.2% **(+19%)**
- Refined accuracy: 89.2% **(+19%)**

Summary

Custom accuracy vs ReFinED metrics:

- ***custom accuracy*** = measures row-level correctness
    - only checks whether first entity each row is predicted correctly
    - Uses whole row for context for prediction, but ignores other entities in row
    - Makes it stricter, coarser metric

- ReFinED metrics = measures entity-level correctness
    - every predicted entity in the row is compared to its gold entity, and metrics like precision, recall, F1 count all entities, not just first.
    - Gives more fine-grained and complete assessment of models overall performance

Merged datasets improve cross-domain performance:

- Small merged datasets (10k) already improve both companies and movies accuracy
- Larger datasets (60k and full 135k) give incremental gains while keeping balanced performance.

Domain-specific full datasets excel in own domain, but hurt other:

- 'companies_full' gives a 7% increase in companies, but movies drop (-31.9%)
- 'movies_full' gives a 19% increase in movies, but companies drop (-64%)

Merged full dataset (135k) is the best overall model:

- Custom accuracy for main entity each row shows a decent score with 70-90% accuracy
    - Companies: +6.6% row-level accuracy
    - Movies: +19% row-level accuracy
- ReFinED evaluation confirms strong entity-level performance across both domains.
    - strong to near perfect precision for both movies and companies

Training considerations:

- Beyond ~60k lines, returns diminish significantly
- Merged dataset recommended for balanced, ***multi-domain performance***
- Single-domain datasets are best for ***domain-specific applications***